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SKELETON-BASED ANOMALY DETECTION IN CITATION NETWORKS

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MOTIVATION

Dynamic Nature
Citation networks evolve constantly, static models fail to capture this dynamics.

Complexity
Hierarchical structures and complex behaviors require continuous-time modeling.

Challenge
Existing methods miss temporal evolution.

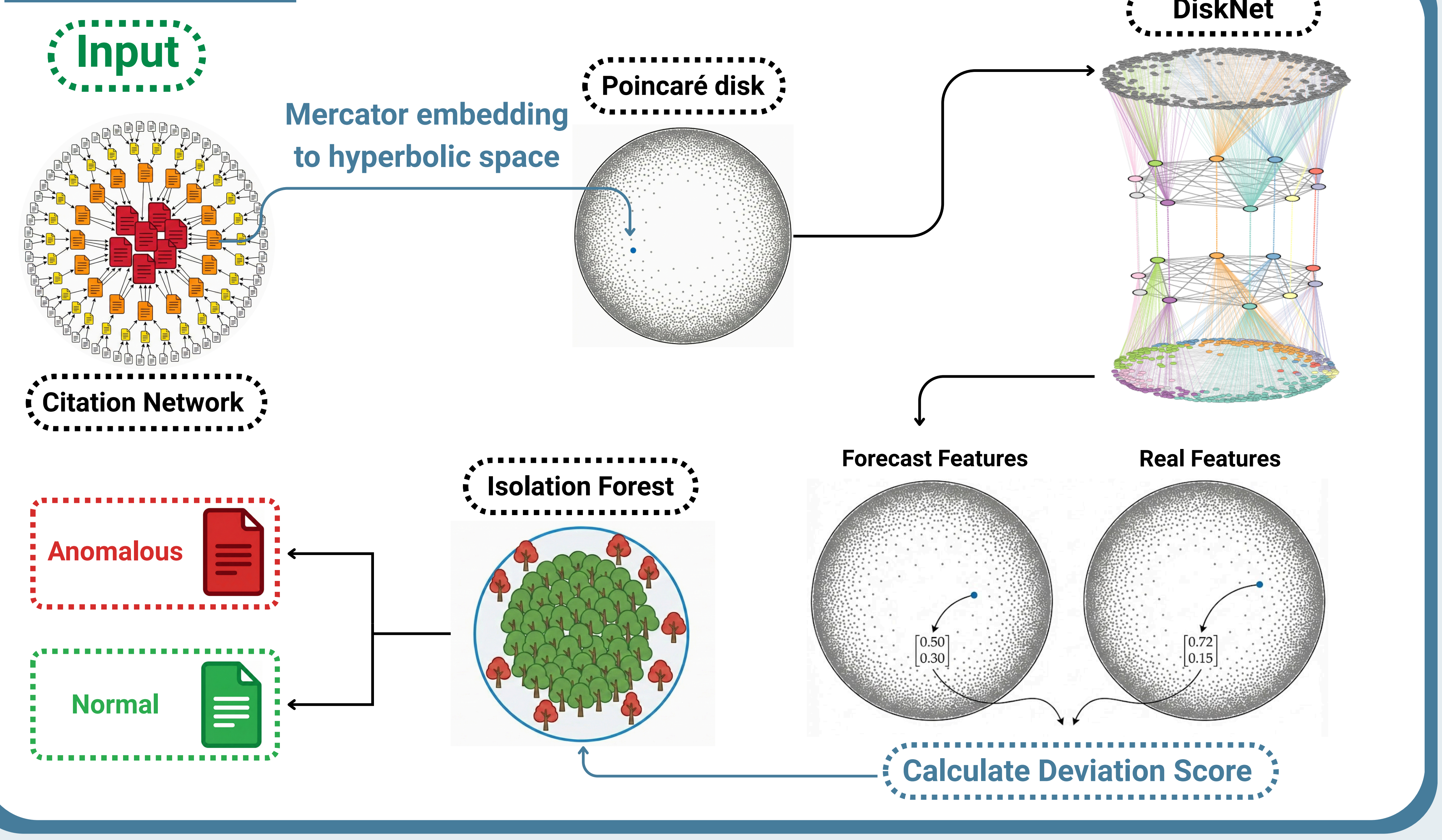
PROBLEM RESOLUTION

Goal
Predict future citation behaviors and detect structural anomalies.

Method
A geometry-aware framework combining Hyperbolic Embeddings with Neural ODEs.

Detection
Identify nodes deviating from their predicted evolutionary trajectory.

WORKFLOW



DISKNET

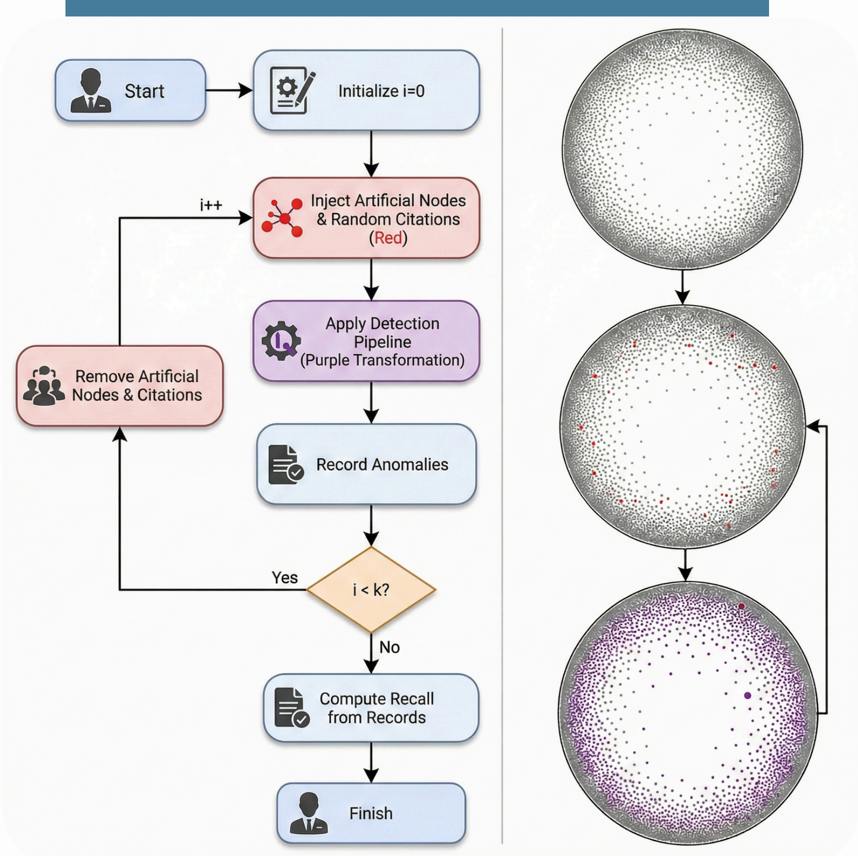
Hyperbolic Geometry
Utilizes the Poincaré disk model, as its negative curvature naturally accommodates the exponential growth and hierarchical topology of citation networks.

Skeleton Extraction
Aggregating geometrically similar nodes into "Supernodes" to reduce dimensionality and noise.

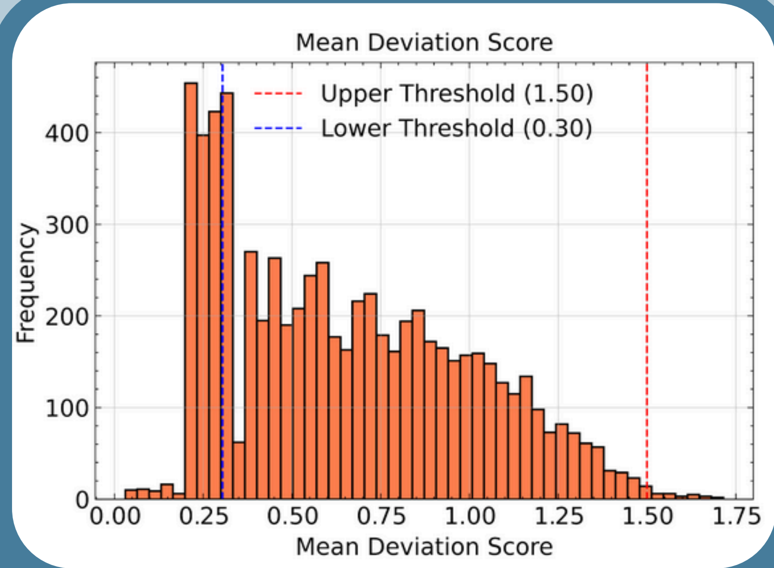
Neural ODEs
Modeling the continuous-time evolution of the "skeleton" in the latent space.

Lifting & Refinement
Projecting coarse skeleton dynamics back to fine-grained node trajectories for anomaly scoring.

VALIDATION PROCCES



RESULTS



The Figure reveals the separation between normal and anomalous nodes. The legitimate papers are clumped between the thresholds

Hyperbolic embedding
A dense core of standard papers contrasts with distinct peripheral anomalies.

Detection Results	Avg (10 Iterations)
Original Anomalies	1020
Original Anomalies (%)	16.32
Synthetic Anomalies	453
Synthetic Anomalies (%)	**72.74**

Consistency: The model demonstrated exceptional stability across 10 independent iterations, achieving a consistent average Recall of ~72.7%.

IMPLEMENTATION TECHNOLOGIES

- Development Environment
- Optimize Training
- Isolation Forest
- Anomaly Detection
- Visualization

CONCLUSIONS

- Citation features are a primary driver of embedding performance.
- Provides a scalable solution for maintaining academic database integrity.
- Structural anomalies are detectable via latent hyperbolic geometry deviations.